Regular cone evolution and local symplectic flux in an Einstein–scalar shell model

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Abstract

A singular radial master variable can obscure regular evolution across a characteristic cone. For the nonconstant regular branch of a fixed five-dimensional Einstein–scalar shell model, we derive an equivalent ambient master equation with smooth coefficients and Minkowski principal symbol. An explicit local differential reconstruction gives metric and scalar perturbations smooth at the cone and its vertex. Direct field equation identities prove sufficiency without division by a tangential harmonic eigenvalue. Standard normally hyperbolic Cauchy theory then constructs actual coupled perturbations from smooth compact data. Before their support reaches the shell, either reflection parity satisfies the homogeneous junction conditions with zero shell displacement. An exact angular-harmonic Cauchy-jet construction gives two non-gauge solutions with nonzero local Einstein–scalar null symplectic flux. This closes a local solution-existence gap in the preceding conditional flux result. The physical Cauchy symplectic form is also nondegenerate on the image of smooth master data supported in a sufficiently small spatial ball. Evolution after shell contact, global phase-space completeness, nonlinear stability and cosmological interpretation remain open.

Status. Reproducible technical continuation, not an APS submission or external peer review. The novelty comparison is bounded; no claim of worldwide priority or evidence for a higher-dimensional Big Bang is made.

1. Question and fixed background

The preceding checkpoint [1] derived the actual gravitational–scalar symplectic current and its cone limit, conditional on actual solutions having differentiable characteristic expansions. Its explicit nonzero-flux example was a separate flat test-scalar problem. Here we construct a family of solutions of the registered coupled equations that realizes nonzero flux. The result is local in ambient time and uses finite propagation to meet the shell conditions before contact. It does not solve the subsequent initial-boundary-value problem.

Use signature $(-++++)$, $8\pi G_5 = 1$, and the doubled-interior action

$$S = \sum_{i=\pm} \left\{ \int_{M_i} \epsilon_g [R/2 - (\nabla\phi)^2/2 - U(\phi)] + \int_\Sigma \epsilon_h K_i \right\} - \int_\Sigma \epsilon_h \sigma(\phi). \tag{1}$$

Both outward shell normals point toward increasing interior radius. The registered functions and detuning interval are

$$\begin{gathered} \eta = \phi - 1, \quad W = \tfrac{1}{3} + \eta^2 + \tfrac{1}{3}\eta^3, \quad U = \tfrac{1}{2}W_{,\phi}^2 - \tfrac{2}{3}W^2, \\ \sigma = 2W + \delta(1 + c + c\eta), \quad c = \frac{2}{1.0357712571566784} - \frac{4}{3}, \quad 0 < \delta \le 0.002. \end{gathered} \tag{2}$$

The decimal is exact. Fix one background on the already certified regular branch [2]; this note does not recertify that branch or take a uniform $\delta \to 0$ limit. Its warped form and equations are

$$g = dy^2 + \rho(y)^2\gamma, \quad \mathrm{Ric}\,\gamma = 3\gamma, \quad a = \rho'/\rho, \quad p = \phi',$$

$$p' = U_{,\phi} - 4ap, \quad a' = -\rho^{-2} - p^2/3, \quad a^2 = \rho^{-2} + p^2/12 - U/6, \qquad a_b = \sigma_b/6, \quad p_b = -\sigma_{,\phi,b}/2. \tag{3}$$

The regular endpoint has $\rho = y + O(y^3)$, $p = \kappa y + O(y^3)$, $\kappa \neq 0$, with analytic coefficients in y^2. On this fixed branch $p < 0$ for $0 < y \leq T_b$. Analyticity is used to extend the background across the cone. The perturbations themselves need only be smooth.

Introduce the exact conformal radius by

$$\frac{dr}{r} = \frac{dy}{\rho}, \quad r/y \longrightarrow 1, \quad s = r^2 = X \cdot X, \quad g_{AB} = \Omega(s)^2 \eta_{AB}, \quad \rho = r\Omega(s), \quad \phi = \Phi(s), \quad f = \Phi_s. \tag{4}$$

Here $X = (T, \mathbf{X})$, $R = |\mathbf{X}|$, $s = R^2 - T^2$, and $\Omega(0) = 1$, $f(0) = \kappa/2 \neq 0$. Analytic continuation gives a neighborhood $-\epsilon < s < s_+$ with $\Omega > 0$, $f \neq 0$; the background equations continue there. This is a local Lorentzian extension, not a singular creation surface. Define

$$E = X^A \partial_A, \qquad L = \frac{\Omega_s}{\Omega} + 2\frac{f_s}{f}, \qquad \mathcal{L} = \Box_\eta - 2L(s)E - \left[8L(s) + \frac{8}{3} s f(s)^2\right]. \tag{5}$$

2. Exact regular master and reconstruction

In the longitudinal scalar slice write $h_{yy} = -4\Psi$, $h_{y\mu} = 0$, $h_{\mu\nu} = 2\rho^2 \Psi \gamma_{\mu\nu}$, and $\delta\phi = \chi$. The inherited unreduced master is

$$P = \rho^2 \Psi, \quad \chi = -\frac{3P_y}{\rho^2 p}, \qquad M[P] = P_{yy} - (2a + 2p'/p)P_y + \rho^{-2}(\Box_\gamma + 4)P - \frac{2}{3}p^2 P. \tag{6}$$

The expression appears singular because p vanishes at the endpoint. Set

$$P = r^4 F(X). \tag{7}$$

Using $\partial_y = \Omega^{-1}\partial_r$, $p = 2rf/\Omega$ and $\Box_\eta = \partial_r^2 + 4\partial_r/r + \Box_\gamma/r^2$ on $s > 0$ gives the exact identity

$$M[r^4 F] = \Omega^{-2} r^4 \mathcal{L} F. \tag{8}$$

The reconstructed fields take the manifestly regular ambient form

$$\boxed{h_{AB} = 2sF\eta_{AB} - 6FX_A X_B, \qquad \chi = -\frac{3}{2\Omega^2 f}(E + 4)F.} \tag{9}$$

Indices on X in this expression are lowered with η. For example, $dy^2 = \Omega^2 (X_A dX^A)^2/s$; cancellation with $\Psi = sF/\Omega^2$ removes its apparent pole. Equation (9) is smooth even at $X = 0$.

Proposition 1 (sufficient coupled solution family). Every smooth solution F of $\mathcal{L}F = 0$ in the extended background reconstructs through (9) a solution of all linearized bulk Einstein and scalar equations. No completeness assertion is made about arbitrary gravitational perturbations or exceptional scalar harmonic sectors.

Proof. Define the trace-reversed Einstein and scalar residuals by

$$\mathscr{E}_{AB} = R_{AB} - \phi_{,A}\phi_{,B} - \tfrac{2}{3} U g_{AB}, \qquad \mathscr{S} = \Box_g \phi - U_{,\phi}.$$

Direct metric variation, followed only by (3) and the reconstruction, gives

$$\delta\mathscr{E}_{yy} = 2M[P]/\rho^2, \quad \delta\mathscr{E}_{y\mu} = 0, \quad \delta\mathscr{E}_{\mu\nu} = -\gamma_{\mu\nu} M[P],$$

$$\delta\mathscr{S} = -\frac{3}{\rho^2 p}(\partial_y + 2a) M[P]. \tag{10}$$

Equivalently, in ambient coordinates,

$$\boxed{\delta\mathscr{E}_{AB} = \Omega^{-2}(3X_A X_B - s\eta_{AB})\mathcal{L}F, \qquad \delta\mathscr{S} = -\frac{3}{2\Omega^4 f}(E + 6)\mathcal{L}F.} \tag{11}$$

These are differential-operator identities. Their coefficients extend analytically in s, so identities established for $s > 0$ continue across $s = 0$ and hold for arbitrary smooth test F. All residuals vanish when $\mathcal{L}F = 0$. No division by $\Box_\gamma + 4$ or by a separated eigenvalue occurs. The accompanying direct-index checker verifies all 15 independent Ricci components and the scalar variation; a separate ADM calculation checks the constraints and scalar equation. □

3. Cauchy existence, finite energy and the shell

The principal symbol of $\mathcal{L}$ is precisely that of five-dimensional Minkowski space. Its lower-order coefficients are smooth where $f \neq 0$. Standard normally hyperbolic Cauchy theory [3] therefore supplies a unique smooth local solution for any smooth compact initial data (F, F_T) on a spacelike slice $T = t_0$, with propagation speed at most one. One may extend the coefficients smoothly outside a compact causal neighborhood, solve the auxiliary Cauchy problem, and restrict back. Domain of dependence makes the restricted solution independent of that extension.

For an explicit energy estimate, the positive weight $A = (\Omega f^2)^{-1}$ satisfies $A_s/A = -L$. With $\psi = \sqrt{A}F$, equation (5) is equivalent to

$$(\Box_\eta - V_c)\psi = 0, \qquad V_c = 3L - 2sL_s + sL^2 + \tfrac{8}{3}sf^2. \tag{12}$$

On a compact time slab containing the causal support, let $K = \sup|1 - V_c|$. The auxiliary energy

$$\mathcal{E}(T) = \frac{1}{2}\int_{\mathbb{R}^4} (|\psi_T|^2 + |\nabla\psi|^2 + |\psi|^2) d^4X$$

satisfies $|\mathcal{E}'| \leq K\mathcal{E}$, hence $\mathcal{E}(T) \leq \mathcal{E}(t_0)e^{K|T-t_0|}$. Smooth coefficients and commuted estimates preserve smoothness. This is a control norm for the master equation. It is not identified with the gravitational symplectic norm, a conserved physical energy, or a proof of absence of gravitational ghosts. Smooth compact reconstructed fields have finite local norms; derivative reconstruction changes the required Sobolev order.

Proposition 2 (short-time doubled-shell solutions). Suppose the causal support of compact initial data stays within a region of the analytically extended fixed background and remains a positive distance from the shell for $|T - t_0| < \Delta$. Then (9), with either even or odd copying to the second interior and zero shell displacement, solves the full homogeneous linearized bulk and junction equations on that slab.

Proof and usable criterion. The shell is the timelike hyperboloid $R_b(T) = \sqrt{r_b^2 + T^2}$, with $r_b = r(T_b) > 0$. Data supported in $R \leq R_0$ propagate inside $R \leq R_0 + |T - t_0|$. Thus $R_0 + \Delta < r_b$ ensures an open zero-field neighborhood of the shell. All local derivatives of the reconstructed fields vanish there. With zero displacement, every linearized matching residual vanishes, including the scalar junction and the moving-shell terms. The bound $(|t_0| + \Delta)^2 < \epsilon$ keeps the timelike part in the available background extension; alternatively choose the whole support in an even smaller cone neighborhood. The positive-s support must also remain in the actual background interval, for example by $(R_0 + \Delta)^2 < s_+$. The field may be set to zero beyond that neighborhood where its causal support is absent. The construction works for each parity. □

One explicit choice also includes the cone vertex. If the actual analytic coefficient neighborhood contains $|s| < \sigma$, choose $t_0 > 0$ so that $13t_0^2 < \sigma$ and $4t_0 < r_b$, and initial support in $t_0/2 < R < 3t_0/2$. On $-t_0 < T < 3t_0$, its causal support obeys $R \leq 7t_0/2$ and hence $|s| < 13t_0^2$. It remains strictly inside the shell throughout this slab, which contains both the vertex and the finite cone cut used below. Subsequent support reaching $R = 0$ causes no problem for smooth Cartesian Cauchy evolution.

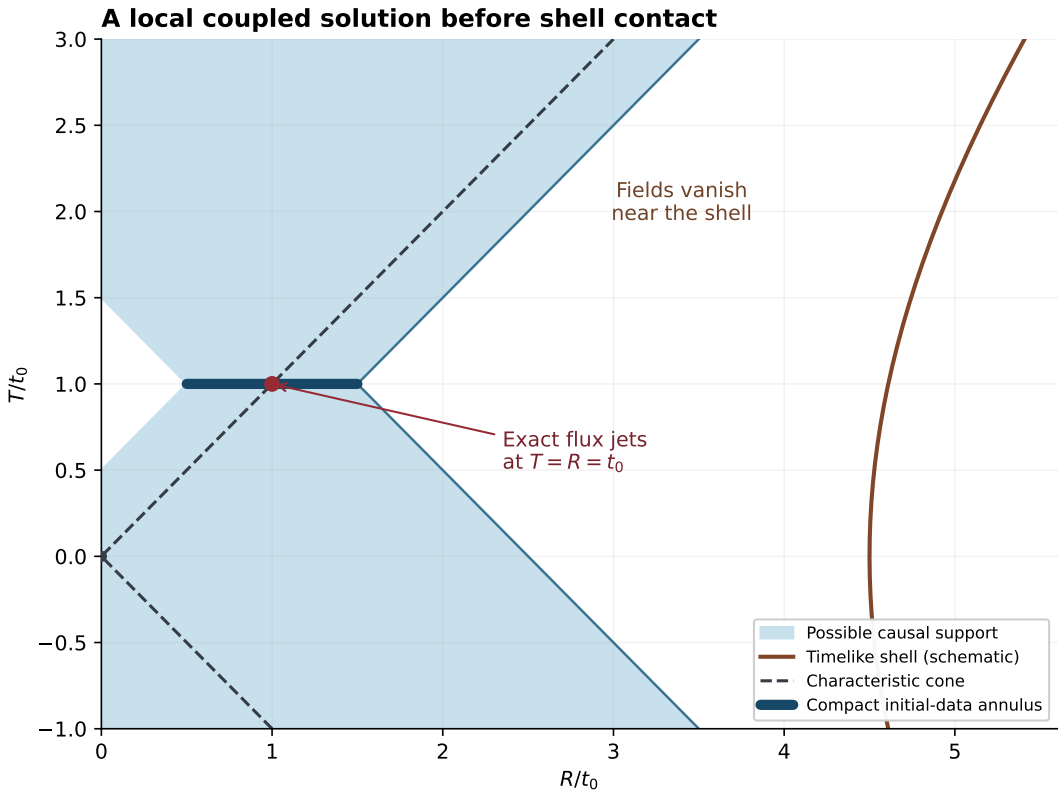


Figure 1: Finite-propagation geometry, in units of the chosen local time scale. The illustrative shell ratio is $r_b/t_0 = 4.5$; this is a geometric schematic, not a numerical background or perturbation solution. The shading bounds possible support, not wave amplitude.

This result deliberately ends before shell contact. It provides no reflection law, all-time Wentzell evolution, boundary compatibility hierarchy after contact, or global Hilbert-space identification. Prior scalar Wentzell results [4] and current vacuum Einstein boundary theorems [5,6] do not by themselves supply the required coupled Israel-shell theorem.

4. Actual non-gauge solutions with nonzero cone flux

Let $u = R - T$, $v = R + T$, so $s = uv$. Work on a finite future-cone segment $u = 0$, $v > 0$, away from its vertex. Smoothness of F in (9) supplies the uniform differentiated expansions required in [1]. Its limiting scalar is

$$\chi_0(v, \vartheta) = -C(4 + v\partial_v)F_0(v, \vartheta), \qquad C = \frac{3}{2f(0)} = \frac{3}{\kappa}. \tag{13}$$

For the convention of increasing radial normal, the per-copy current density tends to

$$\frac{v^3}{8}(\chi_{01}\partial_v\chi_{02} - \chi_{02}\partial_v\chi_{01})\, dv\, d\Omega_3. \tag{14}$$

Taking both solutions with the same reflection parity, either even or odd, gives twice (14); a mixed-parity pair cancels between copies. This is the actual Einstein–scalar symplectic current restricted to this solution family, not the auxiliary current of (12). It is antisymmetric in the two solutions and is not a positive energy flux.

We now construct its nonzero value directly from freely specified Cauchy data, without assuming a characteristic boundary-value theorem. Fix small $t_0 > 0$, and let H_l be a real unit-normalized spherical harmonic on S^3, $\Delta_{S^3}H_l = -\lambda H_l$, $\lambda = l(l+2)$, $l \geq 2$. Take a smooth radial cutoff $\zeta(R)$ supported in an annulus about $R = t_0$

and equal to one near that sphere. Specify at $T = t_0$

$$F_1 = a\zeta H_l, \quad F_{1T} = 0, \qquad F_2 = 0, \quad F_{2T} = b\zeta H_l, \qquad ab > 0. \tag{15}$$

The annulus makes these smooth compact Cartesian data. Rotational invariance preserves l; $l \geq 2$ also avoids the constant and Hessian-degenerate harmonic subspaces relevant to the earlier scalar reduction. Choose its support and the slab to satisfy Proposition 2.

At $T = R = t_0$ ($v_0 = 2t_0$), put $L_0 = L(0)$. Equation (5) fixes the second time derivatives:

$$F_{1TT} = -(\lambda/t_0^2 + 8L_0)aH_l, \qquad F_{2TT} = -2L_0 t_0 bH_l.$$

Since $\partial_v = (\partial_T + \partial_R)/2$ and $\partial_v s = 0$ on the cone, the reconstructed jets are exactly

$$\begin{aligned} \chi_1 &= -4CaH_l, \quad \chi_{1v} = Ca[\lambda/(2t_0) + 4t_0 L_0]H_l, \\ \chi_2 &= -Ct_0 bH_l, \quad \chi_{2v} = -Cb[5/2 - t_0^2 L_0]H_l. \end{aligned} \tag{16}$$

Consequently,

$$\boxed{\chi_1\chi_{2v} - \chi_2\chi_{1v} = C^2 ab(10 + \lambda/2)H_l^2.} \tag{17}$$

The background-dependent terms cancel. For $l = 2$, the coefficient is $14C^2ab$. After angular integration the doubled current density at v_0 is $28t_0^3C^2ab > 0$. Continuity makes the integrated flux strictly nonzero on a sufficiently short v interval around v_0. Its sign can be reversed by exchanging the two solutions, as a symplectic bilinear must.

Nonzero current alone is not a gauge-invariance test at a null boundary. For a separate physical witness, use the background scalar $K_g = (\nabla\phi)^2 = 4sf^2/\Omega^2$ and its relational perturbation

$$Z = \delta K_g - \frac{(K_g)_s}{f}\chi. \tag{18}$$

It is invariant under smooth infinitesimal diffeomorphisms: the two background Lie derivatives cancel. On the cone it equals $Z = 4f(0)(v\chi_v - \chi)$. For (16),

$$\begin{aligned} Z_1 &= 4f(0)Ca[\lambda + 4 + 8t_0^2 L_0]H_l, \\ Z_2 &= 4f(0)Ct_0 b[-4 + 2t_0^2 L_0]H_l. \end{aligned} \tag{19}$$

Choosing $l = 2$ and $|t_0^2 L_0| < 1$ makes both nonzero functions. Thus each solution is non-pure-gauge under smooth diffeomorphisms. This does not classify finite-cut gauge charges or prove that the current descends to a complete global quotient.

5. Prior work, verification and remaining questions

Regular normally hyperbolic Cauchy evolution is established mathematics [3]. Gauge-invariant scalar master variables in gravitating walls and brane stabilization have substantial prior literature [7,8]. Their variables and zero-set hypotheses must be compared before using them at the cone. In particular, a variable containing $1/\phi'$ is not made regular merely by calling it gauge invariant. The present calculation supplies an explicit transformation, smooth reconstruction and full residual factorization for the fixed branch (2).

Earlier private HDBLAST notes M465R1 and M471 already discussed regular ambient evolution and cone transparency for a branch near $\phi = -1$. Their conceptual observation is acknowledged. Their negative-branch numerical focusing bounds and stationary transfer conclusion do not transfer to the present branch near $\phi = +1$. The added result here is the specific regular master, reconstruction, exact non-gauge flux pair and finite-time shell argument for the registered current model. The accompanying literature register records the bounded searches, selected primary passages, and exclusions; it does not establish external originality.

The capsule includes separate direct-index and ADM derivations, exact symbolic checks, deliberately wrong formulas, normal and optimized Python reproduction, and integrity verification. These checks support the

stated algebra. The analytic Cauchy theorem, support argument and proof hypotheses are necessary parts of the result; check counts are not counts of independent physical discoveries. No numerical evolution or observational fit is presented.

The next decisive task is to evolve these smooth data through first shell contact with the correct coupled junction conditions and energy estimate. A subsequent task is to identify the allowed gauge quotient and compare its physical phase space with the earlier radial spectral closure. Neither the existence result here nor the positive radial theorem alone settles that comparison. Nonlinear evolution, reheating, cosmological observables and discrimination from four-dimensional rivals remain open. The proposed higher-dimensional origin of the Big Bang remains unestablished.

Appendix A. Exact boundary laws and a local Cauchy form

The supplemental boundary audit supplies two further precise statements. At the fixed shell define $B_b = p_b'/p_b + \sigma_b''/2 > 0$. Transforming the inherited master junctions by $P = r^4 F$ gives

$$nF - B_b^{-1}\Box_{h_b}F + \left(\frac{4}{\rho_b} - \frac{4}{\rho_b^2 B_b}\right)F = 0 \quad \text{(even)},$$

$$nF + \left(\frac{4}{\rho_b} + \frac{p_b^2}{3a_b}\right)F = 0 \quad \text{(odd)}, \tag{20}$$

where $n = \partial_y$ is the outward physical normal. The first equation has a tangential wave operator (a Wentzell condition); the second is Robin. Their exact conversion is useful for the next evolution problem. It does not prove that problem well posed after contact, and the positive boundary kinetic coefficient of an auxiliary master action is not identified with the physical Einstein–scalar boundary kinetic form.

For a separate bounded phase-space result, take smooth real compact master data $(q, p) = (F, F_T)|_{T=0}$ in a sufficiently small ball. Here p denotes initial velocity only, not the background radial derivative in (3). Put $D_r = r\partial_r$ and $m_r = 9r^3/(4\Omega f^2)$. The actual per-copy Einstein–scalar current, with future Cauchy orientation, pulls back to

$$\sigma((q_1, p_1), (q_2, p_2)) = K(q_1, p_2) - K(q_2, p_1),$$

$$K(q, p) = \int dr\, d\Omega_3 \left[m_r(D_r + 4)q(D_r + 5)p + \frac{6r^7}{\Omega}qp\right]. \tag{21}$$

Indeed $\chi = -3(D_r + 4)q/(2\Omega^2 f)$ and $\chi_T = -3(D_r + 5)p/(2\Omega^2 f)$ on this slice; the extra unit follows from $[\partial_T, E] = \partial_T$. Transforming the physical time current gives the measure $\Omega^3 r^3$. Compact-support integration by parts, using $r\partial_r \log m_r = 3 - 2sL$, yields

$$K(q, q) = \int dr\, d\Omega_3 \left[m_r\{(D_r q)^2 + (2 + 9sL)q^2\} + \frac{6r^7}{\Omega}q^2\right]. \tag{22}$$

Choose the ball so that $2 + 9sL \geq 1$. Then $K(q, q) > 0$ for every nonzero q. A pair in the kernel of σ against all compact test pairs must have $q = 0$ by testing against $(0, q)$, and then $p = 0$ by testing against $(p, 0)$. Thus this physical pullback is nondegenerate on the stated smooth compact master-data image. K itself need not be symmetric. The result does not identify a completed Hilbert space, a global gravitational Hamiltonian, or the full gauge quotient. The shell contribution vanishes here because the data have support away from the shell.

Declarations and availability

Ricardo Maldonado, Independent Researcher; ORCID 0009-0009-3937-6527. No funding. No known competing interests. The earlier author declaration states that the manuscript is not under consideration elsewhere. No new APS submission accompanies this continuation, and the prior DX13808 rejection is not reversed by these calculations.

Substantial OpenAI Codex assistance was used for mathematical derivation, symbolic programming, literature organization, internal review and drafting. Separate agent processes supplied internal checks; they are not external human peer review. The author retains responsibility for review and any submission. Source, code, exact receipts and this draft are supplied in the associated GitHub research capsule; the preceding spectral edition is archived at DOI 10.5281/zenodo.23335485 within concept DOI 10.5281/zenodo.17088132.

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